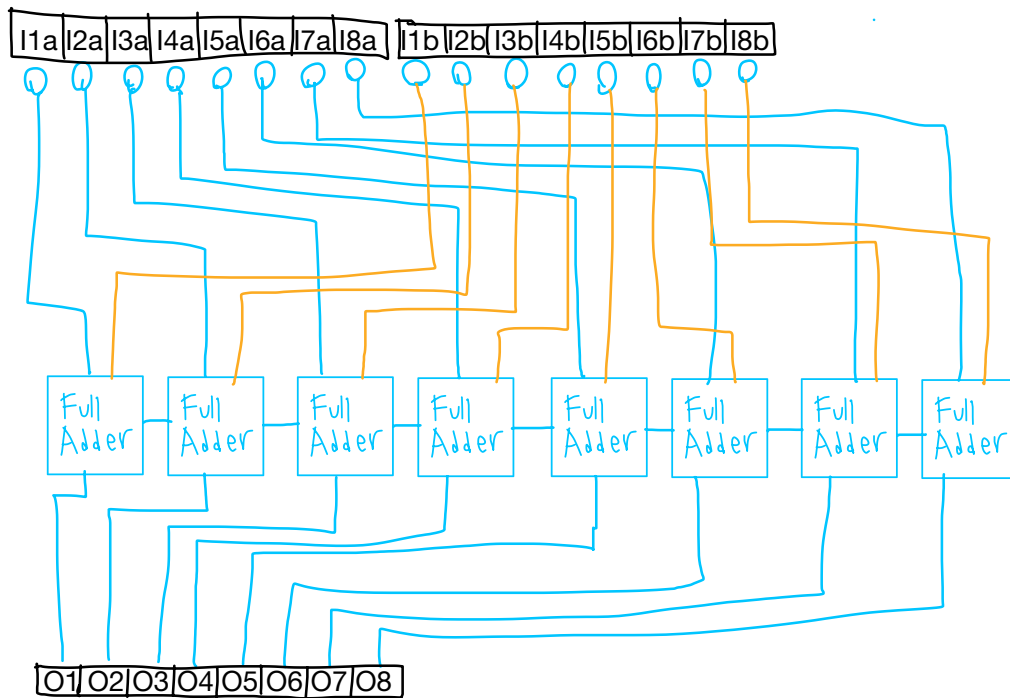




Full Adder

A	B	C _i	Y	C _o
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

	Y	C _i	C _o
AB	1	0	
$\bar{A}\bar{B}$	0	1	
$\bar{A}B$	1	0	
$A\bar{B}$	0	1	

$$Y = C_i(A\bar{B} + \bar{A}B) + C_i(\bar{A}B + AB)$$



	C _o	C _i	C _o
AB	1	0	
$\bar{A}\bar{B}$	0	1	
$\bar{A}B$	0	0	
$A\bar{B}$	0	0	

$$C_o = AC_i + BC_i + AB$$

$$C_o = C_i(A+B) + AB$$

00
01
10
11

could use xor output here

This is the truth table and circuit diagram for 1 full adder. They all are the exact same. The C_o of one is connected to the C_i of the next.

Starting with the first full adder, the 1st digit of each binary number is used. The next full adder uses the 2nd digit of each binary number and so on.

A = input digit 1

B = input digit 2

C_i = input carry

Y = Output "added" digits

C_o = Output carry